2024, Mar. 2024 Release LOC:Acute Pediatric
Bowel Obstruction

Utilization Benchmarks

Length of Stay: **4.5 d**
Length of Stay Type: **InterQual**
Condition/Procedure: **BOWEL OBSTRUCTION**
% Paid as Observation: **13%**

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Overview

Instruction:
This subset is intended to manage children or adolescents with a partial obstruction of the small or large intestine undergoing medical management only and does not address ileus or pseudo-obstruction. Strangulation (with compromised blood flow), complete obstruction, volvulus, or intussusception are considered surgical emergencies. For patients requiring surgical intervention, refer to the General Surgical (Pediatric) subset.

Introduction:
Bowel obstruction is caused by a variety of processes in children and adolescents. The most common causes are intussusception, hernia, malrotation with midgut volvulus, congenital defects, and postoperative adhesions. Typical symptoms of bowel obstruction are irritability (in small children) with nausea, vomiting, abdominal pain, distention, and decreased or no bowel movements. Clinical presentation varies and no one symptom on its own is sufficient to identify bowel obstruction. Abdominal pain associated with bowel obstruction in children is often described as crampy and episodic, persisting for a few minutes at a time. If there is suspicion of bowel obstruction, early identification and intervention is paramount to avoid the potential for bowel ischemia or perforation. Like adults, risk factors associated with bowel obstruction include prior pelvic or abdominal surgery, history of malignancy with prior radiation therapy, and inflammatory bowel disease (IBD) (Lautz and Barsness, Semin Pediatr Surg 2014, 23: 349-52). The presence of fever and tachycardia suggest ischemia.

Evaluation and Treatment:
Initial diagnosis is based on thorough medical history, physical examination, and imaging studies (e.g., abdominal x-ray (KUB), computed tomography (CT) scan, magnetic resonance imaging (MRI), ultrasonography). Initial treatment consists of aggressive fluid resuscitation, bowel decompression, early surgical consultation, and administration of analgesia, antiemetics, and antibiotics if indicated. Decompression can be performed by placement of a nasogastric (NG) tube for suctioning and aspiration prevention. In the absence of acute signs of deterioration, observation and conservative treatment is warranted. One systematic review showed the range of observation from 3 to 6.5 days, while another study found that the younger the child, the shorter the observation period, usually only 1 day when the child was under 1

year of age (Lautz and Barsness, Semin Pediatr Surg 2014, 23: 349-52; Lin et al., BMJ open 2014, 4: e005789).
References: (Ten Broek et al., World J Emerg Surg 2018; 13: 24; Reddy and Cappell, Current gastroenterology reports 2017, 19: 28; Azagury et al., Journal of Trauma and Acute Care Surgery 2015, 79: 661-8; Maung et al., Journal of Trauma and Acute Care Surgery 2012, 73: S362-S9)

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, **One:**

- **Episode Day 1, One:** (1, 2)
  (Symptom or finding within 24h)
  - **ACUTE, One:** (3)
    - Bowel obstruction confirmed by imaging and, **Both:** (4, 5)
      - NPO or NG tube to suction (6)
      - IV fluid (7, 8)
- **Episode Day 2, One:** (2, 9)
  - **ACUTE, One:**
    - **Responder,** discharge expected today if clinically stable last 24h, **One:** (10)
      - Clinical stability, **All:**
        - Bowel sounds present
        - Passing flatus or stool
        - Tolerating PO or nutritional route established (11, 12)
    - **Partial responder,** not clinically stable for discharge and requires continued stay, **One:**
      - Surgical intervention required (use **General Surgical** subset)
      - Bowel obstruction, unresolved, **Both:**
        - Finding, ≥ **One:** (13)
          - Abdominal pain or distention
          - Absence of bowel sounds or activity
          - Absence of flatus or stool
          - Continued evidence of obstruction on imaging (5)
          - Vomiting or high NG tube output
        - Intervention, ≥ **One:**
          - IV fluid (7, 8)
          - Parenteral nutrition
      - Bowel obstruction, resolved and initiating PO or tube feeding ≤ 2d
- **Episode Day 3-5, One:** (2, 9)
  - **ACUTE, One:**
    - **Responder,** discharge expected today if clinically stable last 24h, **One:** (10)
      - Clinical stability, **All:**
        - Bowel sounds present
        - Passing flatus or stool
        - Tolerating PO or nutritional route established (12)
    - **Partial responder,** not clinically stable for discharge and requires continued stay, **One:**
      - Surgical intervention required (use **General Surgical** subset)
      - Bowel obstruction, unresolved, **Both:**
        - Finding, ≥ **One:** (13)
          - Abdominal pain or distention
          - Absence of bowel sounds or activity
          - Absence of flatus or stool
          - Continued evidence of obstruction on imaging (5)
          - Vomiting or high NG tube output
        - Intervention, ≥ **One:**
          - IV fluid (7, 8)
          - Parenteral nutrition
      - Bowel obstruction, resolved and initiating PO or tube feeding ≤ 2d
- **Episode Day 6, One:**
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 24h, **One:** (10)
      - Clinical stability, **All:**
        - Bowel sounds present

- Passing flatus or stool
- Tolerating PO or nutritional route established [12]

⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
- **Use Extended Stay subset**
- Refer for secondary review

Discharge, Level of Care, DISCHARGE SCREENS, One: [14]

⊖ **HOME, Both:**
- ⊖ Level of care appropriateness, **Both:**
  - Home environment safe and accessible [15]
  - Patient or caregiver demonstrates ability to manage care
- ⊖ Bowel obstruction discharge planning, **All:**
  - Provide comprehensive written discharge and teaching instructions [16]
  - ⊖ Education on the importance of, **All:**
    - Increasing fluid intake
    - Notifying healthcare provider of pain, fever, chills, nausea, vomiting, blood in stool, recurrence of symptoms, or distention and inability to pass flatus or stool
    - Strictly adhering to discharge dietary recommendations
  - Medication reconciliation [17]
  - Follow-up care planned [18]
  - Identify and address transportation needs

⊖ **HOME CARE, Both:**
- ⊖ Level of care appropriateness, **All:**
  - Home environment safe and accessible [15]
  - Patient or caregiver willing and able to learn care
  - Outpatient management contraindicated or unavailable [19, 20]
  - ⊖ Skilled services, ≥ **One:** [21]
    - Skilled nursing [22]
    - Skilled therapy, PT, OT, or ST
    - Behavioral health [23]
- ⊖ Bowel obstruction discharge planning, **All:**
  - Provide comprehensive written discharge and teaching instructions [16]
  - ⊖ Education on the importance of, **All:**
    - Increasing intake of fluids
    - Notifying healthcare provider of pain, fever, chills, nausea, vomiting, blood in stool, recurrence of symptoms, or distention and inability to pass flatus or stool
    - Strictly adhering to discharge dietary recommendations
  - Medication reconciliation [17]
  - Follow-up care planned [18]
  - Identify and address transportation needs

⊖ **BEHAVIORAL HEALTH, Both:**
- ⊖ Level of care appropriateness, **One:**
  - Support systems available [24]
- ⊖ Skilled treatment, ≥ **One:**
  - Clinical assessment [25]
  - Individual, group, or family counseling
  - Psychiatric, substance, or medication evaluation [26]

⊖ **SKILLED MEDICAL OR THERAPY, Both:**
- ⊖ Level of care appropriateness, **All:**
  - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
  - Treatment precluded in a lower level
  - ⊖ Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h [22]
- Therapy, **All:**
  - Goal directed therapy with rehabilitation potential [27, 28]
  - Functional impairments requiring at least supervision [29]
  - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [17]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [22]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [27, 28]
        - ≥ 2 functional impairments requiring at least minimum assistance [29]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [17]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [27, 28]
      - ≥ 2 functional impairments requiring at least minimum assistance [29]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [17]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [27, 28]
      - ≥ 2 functional impairments requiring at least minimum assistance [29]
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [30]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [17]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
  - Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level

- In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
  - Primary condition or illness and treatment of active comorbid condition(s) (31)
  - Ventilator dependent ≥ 6h/d and weaning planned (32, 33)
  - Acute and comorbid conditions requiring prolonged hospitalization (34)
- Complete prior to facility transfer, **All:**
  - Medication reconciliation (17)
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

Notes:

1:

Expected progress after 24h

- Abdominal pain and distention improving
- Nausea resolving and manageable with medications
- Bowel sounds, bowel movements, and flatus returning
- Vital signs normalizing and within acceptable limits
- Activity increasing
- Preparing for discharge

Consider

- Surgical consultation (e.g., fever, continued or worsening pain, hemodynamic instability, increased nasogastric output, elevated WBC, peritoneal signs)
- Discharge planning

2:

Care Facilitation

- Nutrition counseling should be considered in a clinically stable patient who continues to require supplemental nutrition due to the inability to tolerate a nutritious, solid diet **(Home)**

Potential risk for admission, consider an alternate level of care or readmission program.

- Knowledge deficit or non-adherence with condition management identified **(Home Care, Readmission Reduction Program).** Include patient or caregiver education regarding medication regimen and nutritional teaching.
- Patient who cannot tolerate a solid diet or has gastrointestinal symptoms (e.g., nausea, vomiting) **(Home Care, Readmission Reduction Program)**
- Hospital admission within the last 6 months **(Home Care, Readmission Reduction Team)**
- Diabetes, infection, nutritional deficiencies **(Home Care, Readmission Reduction Team)**

Social Determinants of Health (SDOH)

Social Determinants of Health (SDOH) are conditions in a person's environment that may affect their health. Social needs related to SDOH may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. Children and caregivers should be screened for SDOH-related barriers to health to mitigate any unmet social and psychosocial factors impacting the health of the child and family unit (Kreuter et al., Annu Rev Public Health 2021, 42: 329-44; Sokol et al., Pediatrics 2019, 144:). The following social needs should be assessed on admission to determine which interventions are relevant to the individual child and their family support system. This list may not be all inclusive and all relevant SDOH factors should be considered. Any social need determined to be relevant to a given child and family should be planned for throughout the hospital stay.
*Interventions should be considered for all patients and families if social need is relevant

Alcohol or substance use

- Coordinate care with outpatient primary care or behavioral health providers*
- Refer to addiction social support systems*
- Provide information about community resources and programs for family-centered treatment (e.g., intensive case management, detox, outpatient, residential treatment, day treatment, and medication assisted treatment)
- Provide information about home visitation programs (e.g., Parent child Assistance Program (PCAP) for pregnant and parenting women with substance use disorders aftercare) (Hildebrandt et al., Infant Ment Health J 2020, 41: 677-96; Vanderplasschen et al., Front Psychiatry 2019, 10: 186)
- Provide information about Cognitive Behavioral Therapy (CBT) Multidimensional Family Therapy (MDFT) and/or Pharmacotherapy for youth with co-occurring emotional/mental disorders (Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021)

Child maltreatment, abuse, or intimate partner abuse - sexual and/or psychological abuse

- Arrange for alternative housing situation, foster care, or shelter if home environment is unsafe*
- Assist with creating a long-term safety plan or make appropriate referral*
- Refer to child protective services (CPS) and other domestic violence services if warranted*

- Refer for psychosocial support if warranted*
- Provide community resources and legal information
- Provide information on parent training programs such as the Nurse Family Partnership Program (NFP) (Nurse-Family Partnership National Resources for Families. 2021; Gubbels et al., Int J Environ Res Public Health 2019, 16:)

End of life planning, arrange palliative care or hospice

- Consider palliative care for children with end stage disease
- Consider hospice care for terminally ill children with a life expectancy of less than 6 months

Environmental exposure to toxic substances and other physical hazards

Lead poisoning:

- Provide community resources for primary prevention and secondary prevention strategies (e.g., blood lead level testing and follow up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Tobacco use (e.g., cigarettes, E-cigarettes, vaping):

- Provide education and/or brief counseling about tobacco use cessation and the risk of second-hand smoke (U.S. Department of Health and Human Services, Tobacco Use in Children and Adolescents: Primary Care Interventions. 2021)

Financial resources strained or limited

{tab}Food insecurity:

- Provide information about local food banks and special nutritional programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service Program (Testa and Jackson, Ann Epidemiol 2021, 58: 22-8; Berman et al., Pediatr Rev 2018, 39: 235-46)
- Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

{tab}Income insecurity:

- Identify income resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

Infants and children with Special Health Care Needs

- Provide anticipatory guidance and education to caregivers regarding medical condition and expected progress after discharge over the next few days, months, or years, as warranted (Dosman and Andrews, Paediatr Child Health 2012, 17: 75-80)*
- Refer for follow-up care to neonatologist or other medical subspecialist for infants or children with ongoing and/or chronic medical issues, (e.g., bronchopulmonary dysplasia or feeding dysfunction) (Pediatrics 2008, 122: 1119-26)*
- Refer for adaptive/assistive medical equipment and equipment for technology-dependent children, (e.g., home oxygen, ventilators, monitors, or tube feeding)(Pediatrics 2008, 122: 1119-26)*
- Refer for neurodevelopmental assessment for high-risk infants*
- Refer to post-discharge follow-up programs (e.g., Maternal, Infant, and Early Childhood Home Visiting Program, Early Head Start program for infants or toddlers under the age of 3, and pregnant women in need of medical, mental health, nutrition and education, or the Nurse-Family Partnership for mothers pregnant with their first child and for ongoing nurse care visits through the childs second birthday) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
- Refer for behavioral support and/or counseling for caregivers through therapy groups, telephone support, and/or peer support groups for caregivers of children with similar health needs (Reyes et al., Health Equity 2021, 5: 100-18; Substance Abuse and Mental Health Administration, Treatment

Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021; Pierron et al., BMC Public Health 2018, 18: 1087)
- Refer to lactation consultant in the hospital and home visits for infants at risk for poor lactation or suboptimal infant breast-feeding behavior (Leeman et al., Pediatr Qual Saf 2019, 4: e130)
- Refer for respite services, if warranted

Mental health co-morbidities involved
- Arrange appointment with primary care provider or behavioral health specialist upon discharge*
- Coordinate care with outpatient behavioral health providers*
- Provide information about suicide prevention, including the National Suicide Prevention hotline*
- Refer for psychiatric services for suicide risk screening and intervention
- Refer to school-based mental health interventions, if applicable (Fazel et al., Lancet Psychiatry 2014, 1: 377-87)
- Provide information about Intensive Community-based treatment (e.g., Functional Family Therapy, Intensive Child and Adolescent and Psychiatric Services, Multidimensional Family Therapy, and Intensive Case Management
- Transfer to day treatment program, residential addiction services or partial hospitalization when warranted

Risk for treatment nonadherence
- Provide information about educational and behavioral support initiatives for children at risk of medication nonadherence and their caregivers, such as text message reminders to take medication and electronic medication dispensers
- Provide information about peer support groups for children and/or families with poor insight into illness or noncompliance with treatment (Mental Health America, Peer Support: Research and Reports 2021; National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- Refer to interventions to prevent medical relapse, or readmission, and to enhance medication compliance

Risk for readmission
- Consider transfer to an alternative level of care such as subacute care
- Consider home care services
- Consider early childhood home visiting programs (e.g., Early Head Start; Early On; Nurse Family Partnership; and the Maternal, Infant, and Early Childhood Home Visiting Program (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
- Consider telehealth and web-based training programs (e.g., telepsychiatry or telebehavioral health and mobile health interventions (J Am Acad Child Adolesc Psychiatry 2017, 56: 875-93)
- Consider Intensive in-home behavioral health services
- Provide patient and family with condition-specific information including appropriate planning for prevention of RSV infection and appropriate immunizations

Special Health Risk Behaviors in adolescence and young adulthood
- Arrange follow-up appointment with primary care provider and/or specialists upon discharge*
- Coordinate care with outpatient behavioral health providers*
- Provide information on mentoring programs, (e.g., Big Brothers Big Sisters of America)
- Refer for Intensive Community-Based treatment for children with comorbid behavioral health issues
- Refer to community resources for teen pregnancy prevention/family planning (use of birth control, subsequent pregnancies) (Maness et al., J Adolesc Health 2016, 58: 636-43)

For eating disorders (overweight, obesity, bulimia):
- Provide information on nutrition services including education and resources for diet management, physical activity, and promotion of healthy eating attitudes and behaviors (Gato-Moreno et al., Int J Environ Res Public Health 2021, 18:; Al-Khudairy et al., Cochrane Database Syst Rev 2017, 6: Cd012691)

- Provide information on outpatient psychosocial interventions and Family-Based Treatment (Lock and La Via, J Am Acad Child Adolesc Psychiatry 2015, 54: 412-25)

For parenting resources:

- Provide information on Family Resource Centers and Parenting support groups for young, first-time parents, (e.g., National parent help line) (National Parent and Youth Helpline)
- Provide information on community resources to improve parenting skills (face-to-face or by telephone) (U.S. Department of Health and Human Services, Parenting Resources. 2021)

Unstable living environment or homelessness expected upon discharge

- Assist with health insurance application
- Provide information about homeless shelter application to caregivers if applicable
- Refer to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- Refer to federally sponsored outpatient programs
- Refer to Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community Treatment (Ponka et al., PLoS One 2020, 15: e0230896; Vijverberg et al., BMC Psychiatry 2017, 17: 284)

3:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

4:

Studies recommend intravenous antiemetic administration for reduction of the nausea and vomiting commonly associated with bowel obstruction and to reduce the risk of aspiration. It is suggested that narcotics be avoided, as they slow peristalsis and may mask the symptoms of intestinal perforation or necrosis such as severe pain, rebound tenderness, and abdominal rigidity (Reddy and Cappell, Current gastroenterology reports 2017, 19: 28).

5:

Bowel obstruction is established by history, physical examination, and imaging studies (e.g., abdominal x-ray (KUB), computed tomography (CT) scan, magnetic resonance imaging (MRI), ultrasonography). (Nguyen and Holland, Pediatr Surg Int 2021, 37: 755-63; Chang et al., J Am Coll Radiol 2020, 17: S305-s14).

6:

Current literature suggests patients diagnosed with bowel obstruction be made NPO and have a nasogastric (NG) tube placed to suction to decompress the dilated stomach and intestine, remove any undigested food or accumulating secretions, and prevent emesis or possible aspiration. An increase in NG tube output may indicate complete obstruction and warrants evaluation (Detz et al., Curr Probl Surg 2021, 58: 100893; Ten Broek et al., World J Emerg Surg 2018, 13: 24).

7:

To allow for the advancement of feedings as intravenous (IV) fluids are weaned, the IV fluid rates are ≥ 50%(0.50) of the maintenance fluid requirement. These rates can be calculated using the following (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020):
Weight ≤ 10 *kg* = ≥ 50 *mL/kg/*24h

- Hourly rate = 2.08 (*mL/h*) x kg

Weight > 10-25 *kg* = ≥ 40 *mL/kg/*24h

- Hourly rate = 1.67 (*mL/h*) x kg

Weight > 25-60 *kg* = ≥ 30 *mL/kg*/24h

- Hourly rate = 1.25 (*mL/h*) x kg

Example: An 11 *kg* child is required to receive a minimum of 40 *mL/kg*/24h to meet this criteria point. Multiply 1.67 (*mL*/h) x 11 (child's weight in *kg*) = 18.37 *mL*/h. This is the minimum fluid replacement rate for this child.

8:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

9:

Expected progress after 48-72h

- Abdominal pain and distention resolved
- Nausea resolved
- Bowel sounds, bowel movements, and flatus returning
- Adequate oral nutrition
- Vital signs stable and within acceptable limits
- Activity returned to baseline
- Preparing for discharge

If no, consider

- Surgical consultation
- Nutritional consultation for patients with inadequate oral intake

10:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

11:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (*mL/kg*/d) can be calculated based on current body weight (*kg*):

- 100 *mL/kg*/d for first 10 *kg*
- 50 *mL/kg*/d for second 10 *kg*
- 20 *mL/kg*/d for each additional *kg* in weight

12:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

13:

Symptoms and findings such as increased abdominal pain, distention, tachycardia, fever, and hypotension are worrisome, as they may be indicative of ischemia, necrosis, perforation, or sepsis. It is essential that frequent abdominal re-assessment be performed (Hyak et al., J Surg Res 2019, 243: 384-90; Reddy and Cappell, Current gastroenterology reports 2017, 19: 28).

14:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

15:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

16:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

17:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

18:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

19:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

20:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

21:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

22:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

23:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

24:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

25:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

26:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

27:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

28:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

29:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

30:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

31:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

32:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

33:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6;

Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

34:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.